

CHAPTER 2: DECISION THEORY

Introduction

- Decision theory is an analytic and systematic way to tackle problems
- A good decision is based on logic, considers all available data & possible alternatives, & applies the quantitative approach.
- A bad decision is one that is not based on logic, doesn't use all available info doesn't consider all alternatives & doesn't employ appropriate quantitative techniques

Terminologies:

- Decision Maker (DM) = a person or a group of people who has to make a decision.
- Alternative (alt) = a set of decision/action/strategy to be chosen by DM. DM has a complete control over an alternative.
- State of Nature/ Outcome (SON) = a set of future conditions, an event/outcome over which DM has no control. Example: Market demand, Weather
- Payoff = profit/revenue/benefit or cost resulting from each combination of alt and SON.
- Decision/Payoff table = a table showing the alt, SON and Payoffs.

The Six Steps in Decision Theory (based on Thompson's problem)

- 1) Clearly define the problem at hand
- 2) List the possible alternatives
- 3) Identify the possible outcomes
- 4) List the payoff or profit of each combination of alternatives and outcomes
- 5) Select one of the mathematical decision theory models
- 6) Apply the model and make your decision

Types of Decision-Making Environments

- Type 1: Decision-making under certainty
 - Decision-maker **knows with certainty** the consequences of every alternative or decision choice
- Type 2: Decision-making under risk
 - Decision-maker **does know** the probabilities of the various outcomes
- Type 3: Decision-making under uncertainty
 - Decision-maker **does not know** the probabilities of the various outcomes

Decision Making Under Risk

- Several possible state of nature occur & probabilities are known.
 - Expected monetary value (EMV)
 - Expected value of perfect information (EVPI)
 - Expected opportunity loss (EOL)

Expected monetary value (EMV)

- The weighted sum of possible payoffs for each alternative.
- The expected value or mean value is the long run average value that would result if the decision were repeated a large number of times.

$$\text{EMV (alternative i)} = (\text{payoff of SON1} \times \text{prob of SON1}) + (\text{payoff of SON2} \times \text{prob of SON2}) + \dots + (\text{payoff of last SON} \times \text{prob of last SON})$$

Example 1:

Ismail is considering the purchase of two types of industrial robots. R1 is a large robot capable of performing a variety of tasks, including welding, painting, etc. R2 is smaller and slower robot, but it has all the capabilities of R1. The robots will be used to perform a variety of repair operations on large industrial equipment. Of course, Ismail can always do nothing and not buy any robot. The market for the repair could be either favorable or unfavorable. He has constructed a payoff matrix showing the expected returns of each alternative and the probability of favorable or unfavorable market. The data are presented below:

ALTERNATIVES	STATE OF NATURE	
	FAVORABLE MKT. (RM)	UNFAVORABLE MKT. (RM)
R1	50,000	- 40,000
R2	30,000	- 20,000
DO NOTHING	0	0
PROBABILITIES	0.6	0.4

Choose the best alternative using EMV.

Solution:

$$\text{EMV (R1)} = (50,000)(0.6) + (- 40,000)(0.4) = 14,000$$

$$\text{EMV (R2)} = (30,000)(0.6) + (- 20,000)(0.4) = 10,000$$

$$\text{EMV (DO NOTHING)} = (0)(0.6) + (0)(0.4) = 0$$

Decision: Choose the largest/maximum EMV (RM14,000)

Therefore, Ismail should purchase a large robot, R1

Expected opportunity loss (EOL)

- EOL is the cost of not picking the best solution. Opportunity loss = regret
Minimize EOL = Maximize EMV
- Calculating EOL:
 - Construct OPPORTUNITY LOSS TABLE.
 - Multiply the opportunity loss for each alternative with the probability & add altogether.

OL = optimum/best payoff of a SON – actual payoff received

$$\text{EOL (alternative i)} = (\text{OL of SON1} \times \text{prob of SON1}) + (\text{OL of SON2} \times \text{prob of SON2}) \\ + \dots + (\text{OL of last SON} \times \text{prob of last SON})$$

Example 2:

Using example 1, choose the best alternative using EOL.

Solution:

Optimum/best payoff of favourable market = RM50,000

Optimum/best payoff of unfavourable market = RM0

Opportunity Loss (OL) table:

ALTERNATIVES	STATE OF NATURE	
	FAVORABLE MKT.	UNFAVORABLE MKT.
R1	$50,000 - 50,000 = 0$	$0 - (-40,000) = 40,000$
R2	$50,000 - 30,000 = 20,000$	$0 - (-20,000) = 20,000$
DO NOTHING	$50,000 - 0 = 50,000$	$0 - 0 = 0$
PROBABILITIES	0.6	0.4

$$\text{EOL (R1)} = (0)(0.6) + (40,000)(0.4) = 16,000$$

$$\text{EOL (R2)} = (20,000)(0.6) + (20,000)(0.4) = 20,000$$

$$\text{EOL (DO NOTHING)} = (50,000)(0.6) + (0)(0.4) = 30,000$$

Decision: Choose the smallest/minimum EOL (RM16,000)

Therefore, Ismail should purchase a large robot, R1

Expected value of perfect information (EVPI)

- EVPI is the maximum amount of DM would pay for the perfect information and places an upper bound on what to pay for information.
- $\text{EVPI} = \text{Expected Value with Perfect (EV}_{wPI}) \text{ Information} - \text{maximum EMV} = \text{minimum EOL}$
- To calculate EV_{wPI}

$$\text{EV}_{wPI} = (\text{best payoff of SON1} \times \text{prob of SON1}) + (\text{best payoff of SON2} \times \text{prob of SON2}) + \dots + (\text{best payoff of last SON} \times \text{prob of last SON})$$

Example 3:

Using example 1, what is the maximum amount that Ismail would pay for the perfect information.

Solution:

Optimum/best payoff of favourable market = RM50,000

Optimum/best payoff of unfavourable market = RM0

$$\text{EV}_{wPI} = (50,000)(0.6) + (0)(0.4) = 30,000$$

$$\text{Maximum EMV} = 14,000 \quad (\text{from example 1})$$

$$\text{Therefore} \quad \text{EVPI} = \text{EV}_{wPI} - \text{maximum EMV} = 30,000 - 14,000 = 16,000$$

$$\text{Or} \quad \text{EVPI} = \text{minimum EOL} = 16,000$$

Comment: Based on the EVPI value, the most Ismail would be willing to pay for perfect information is RM16,000.

Example 4:

John Thomson is the founder and president of Thomson Lumber Company, a profitable firm located in Portland, Oregon. The problem that Johnson identifies is whether to expand his product line by manufacturing and marketing a new product, backyard storage sheds. He has several alternatives which he has to decide upon. They are 1) construct a large new plant to manufacture the storage sheds, 2) a small plant, or 3) no plant at all (that is he has the option not to make the new product at all).

Thomson also examines the market in which he is going to sell the new product. He found out that there might be a favourable market where there will be a high demand for the product or there could be an unfavourable market, meaning there is a low demand for the product. He believes that the probability of a favourable market is exactly the same as the probability of unfavourable market. John Thomson has already evaluated the potential profits associated with the various outcomes. With a favourable market he thinks that a large facility would result in the nett profit of RM200,000 to his firm. The conditional value if the market is unfavourable would be RM180,000 net loss. A small plant would result in a net profit of RM100,000 in a favourable market but a net loss of RM20,000 would occur if the market was unfavourable. Finally, doing nothing in RM0 profit in either market.

- Present these values by constructing a decision table, then find out which alternatives would give the highest expected monetary value. Which plant should Thompson build?
- Calculate the EOL for each state of nature. Using the minimum EOL as the decision criterion, what is the best decision?

John Thompson has been approached by Scientific Marketing, Inc., a firm that proposes to help John make the decision about whether to build the plant to produce storage sheds. Scientific Marketing claims that its technical analysis will tell John with certainty whether the market is favorable for his proposed product. In other words, it will change his environment from one of the decision making under risk to one of the decision making under certainty. This information could prevent John from making a very expensive mistake. Scientific Marketing would charge Thompson RM65 000 for the information.

- What would you recommend to John? Should he hire the firm to make the marketing study? Is it worth to pay RM65 000? What is the study worth? Calculate EVPI to get the answers.

Solution:

- Payoff table:

ALTERNATIVES	STATE OF NATURE		EMV
	FAVORABLE MKT.	UNFAVORABLE MKT.	
LARGE	200,000	- 180,000	10,000
SMALL	100,000	- 20,000	40,000
DO NOTHING	0	0	0
PROBABILITIES	0.5	0.5	

$$\text{EMV (LARGE)} = (200,000)(0.5) + (-180,000)(0.5) = 10,000$$

$$\text{EMV (SMALL)} = (100,000)(0.5) + (-20,000)(0.5) = \mathbf{40,000}$$

$$\text{EMV (DO NOTHING)} = (0)(0.5) + (0)(0.5) = 0$$

Decision: Thompson should build a small plant

- b) Optimum/best payoff of favourable market = RM200,000
 Optimum/best payoff of unfavourable market = RM0

OL table:

ALTERNATIVES	STATE OF NATURE	
	FAVORABLE MKT.	UNFAVORABLE MKT.
LARGE	$200,000 - 200,000 = 0$	$0 - (-180,000) = 180,000$
SMALL	$200,000 - 100,000 = 100,000$	$0 - (-20,000) = 20,000$
DO NOTHING	$200,000 - 0 = 200,000$	$0 - 0 = 0$
PROBABILITIES	0.5	0.5

$$\text{EOL (LARGE)} = (0)(0.5) + (180,000)(0.5) = 90,000$$

$$\text{EOL (SMALL)} = (100,000)(0.5) + (20,000)(0.5) = 60,000$$

$$\text{EOL (DO NOTHING)} = (200,000)(0.5) + (0)(0.5) = 100,000$$

Decision: Thompson should build a small plant

- c) $\text{EV}_{\text{wPI}} = (200,000)(0.5) + (0)(0.5) = 100,000$

Maximum EMV = 40,000 (from question a)

$$\text{Therefore } \text{EVPI} = \text{EV}_{\text{wPI}} - \text{maximum EMV} = 100,000 - 40,000 = 60,000$$

$$\text{Or } \text{EVPI} = \text{minimum EOL} = 60,000$$

Recommendation:

Based on the EVPI, the most Thompson would be willing to pay for perfect information is RM60,000. Since the firm charged RM65,000 (more than EVPI), Thompson should not hire the firm for the information.

Decision Tree

- Any problem that can be presented in a decision table can be presented in decision tree.
- Contain Decision Points (*Decision Nodes*) & State of Nature Points (*State of Nature Nodes*).

□ A decision node from which one of several alternatives may be chosen.

○ A state of nature node.

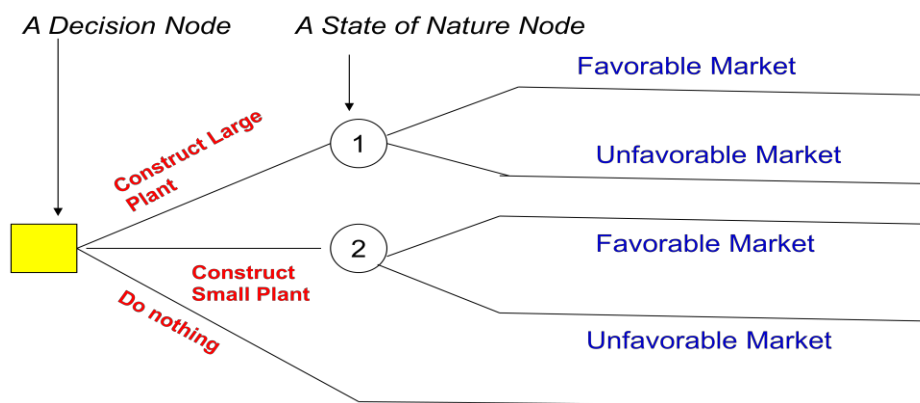
Five Steps of Decision Tree Analysis

1. Define the problem.
2. Structure @ draw the decision tree.
3. Assign probabilities to the state of nature.
4. Estimate payoffs for each possible combination of alternative & SON.
5. Solve the problem by computing EMV for each SON node.

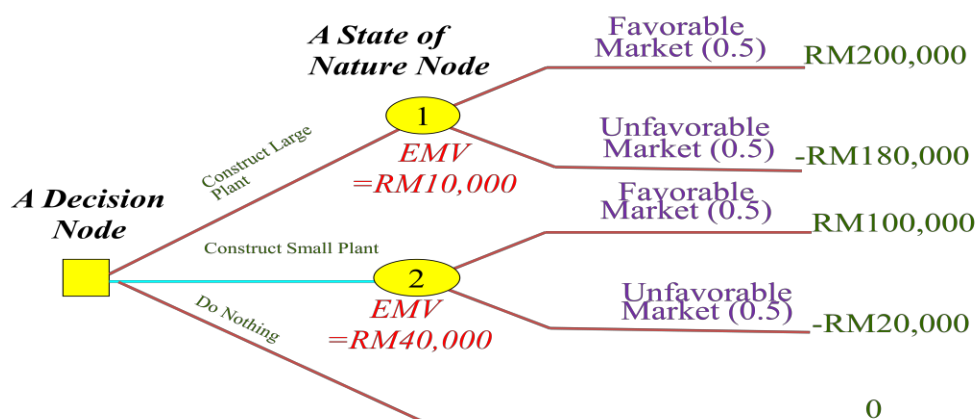
Drawing a Decision Tree

- Begin drawing from the left to the right.
- Lines @ branches from squares represent alternatives.
- Branches from circles represent state of nature.

Thompson's Decision Tree



- Payoffs are placed at the right side of each tree's branches.
- Probabilities are shown in parentheses next to each SON.
- Calculate the EMV for each node.
- The branch leaving the decision node leading to the SON node with the highest EMV will be chosen.



A more complex decision for Thompson Lumber.

- Decision tree are most powerful tool than decision table if sequential decision exist.
- Sequential decision – the 2nd decision depends on outcome of the 1st.

Example 5

Refer to Thompson Lumber with a more complex decision to be made – extra information provided:

Before deciding about building a new plant, John has the option of conducting his own market research survey at a cost of RM10,000. He believes that the probability of favorable market is exactly the same as the probability of unfavorable market. He estimates that the probability of the survey will indicate positive result is 0.45. Furthermore, the revised probability for the favorable market given the survey result is positive is 0.78. The probability that the market is unfavorable given negative survey result is 0.73.

- Construct the complete decision tree with payoffs, probabilities and EMV's for Thompson Lumber.
- Recommend the best decision for Thompson.

HINT**

- Thompson case – 1st : conduct survey, 2nd : construct plant
- Information from survey can help Thompson decide which plant to be built.
- 1st decision point – conduct a RM10000 survey or not.
- If he conducts the survey there will be 2 possibilities that the survey results are favourable (45%) & unfavourable (55%).
- Probabilities are stated in parentheses.
- **Payoffs – the cost of survey must be subtracted from the actual payoffs without survey.**

Solution:

- Cost of the survey = RM10,000

Payoff table: (after subtracted RM10,000)

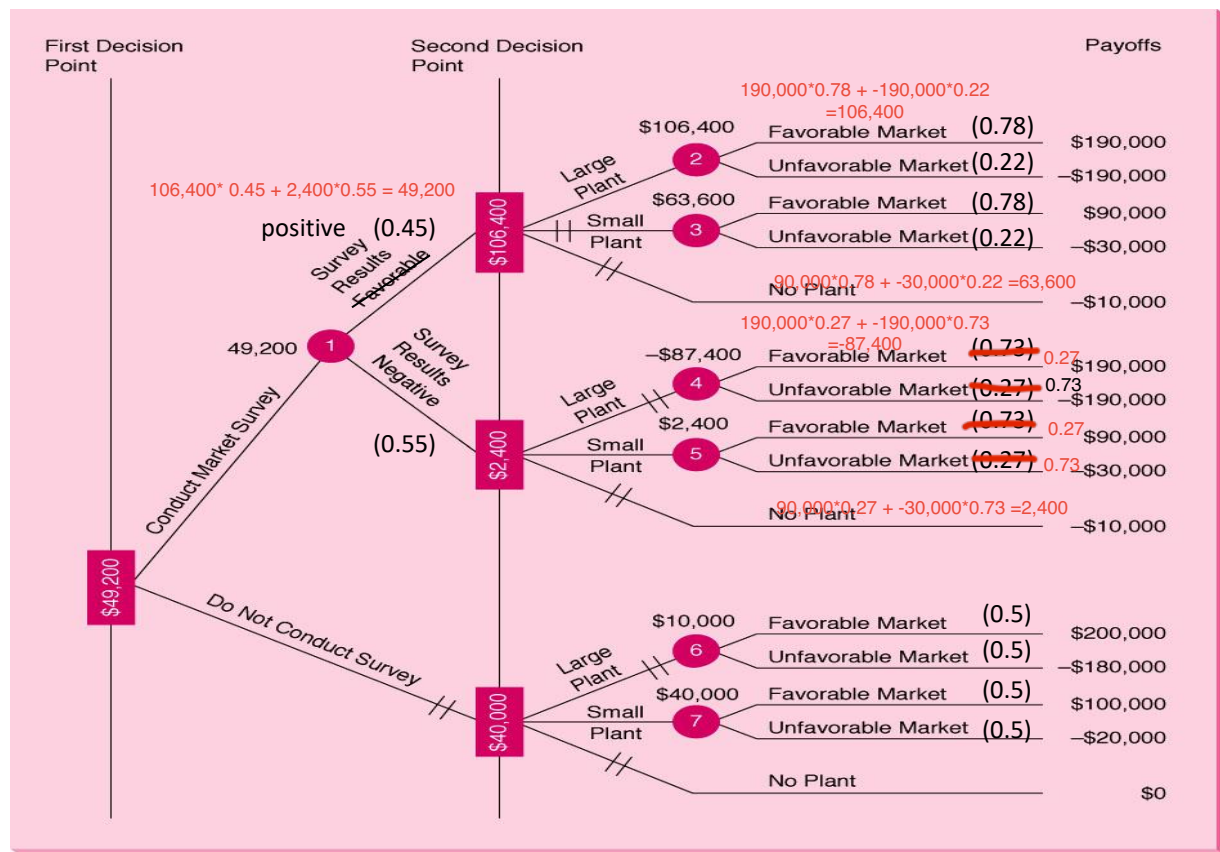
ALTERNATIVES	STATE OF NATURE	
	FAVORABLE MKT.	UNFAVORABLE MKT.
LARGE	$200,000 - 10,000 = \mathbf{190,000}$	$-180,000 - (-10,000) = \mathbf{-190,000}$
SMALL	$100,000 - 10,000 = \mathbf{90,000}$	$-20,000 - (-10,000) = \mathbf{-30,000}$
DO NOTHING	$0 - 10,000 = \mathbf{-10,000}$	$0 - 10,000 = \mathbf{-10,000}$

Let F = favourable + = positive survey result
 U = unfavourable – = negative survey result

$P(F) = 0.5$; $P(U) = 0.5$
 $P(+)= 0.45$; $P(-)=0.55$
 $P(F/+)= 0.78$; $P(U/+)= 0.22$
 $P(F/-)= 0.73$; $P(U/-)= 0.27$

Thompson's Decision Tree

= potong branch jika tidak di pilih .
Hanya pilih yg nilai tinggi



// Alternative is dropped from further consideration because its EMV is lower than the EMV for the best alternative.

- b) **DECISION**
- 1st stage: Conduct survey
 - 2nd stage:
 - i) If survey results favorable (positive): construct large.
 - ii) If survey results unfavorable (negative): construct small.

Expected Value of Sample Information (EVSI)

- Measures the value of sample information (how far the value of survey will help in making decision).
- EVSI is calculated based on imperfect information (SON can't be determined with certainty). The imperfect information is usually obtained from sampling and survey/research and the result obtained from survey will be used to update/improve the information/probability found or estimated
- It can be regarded as the amount the DM is willing to pay for additional information.

$$\text{EVSI} = (\text{EMV with sample info} + \text{cost of survey}) - \text{EMV without sample info}$$

* Note: If payoff values had not already subtracted the study cost, do not add cost of survey.

For Thompson Case:

$$\begin{aligned} \text{EVSI} &= (\text{EMV with sample info} + \text{cost of survey}) - \text{EMV without sample info} \\ &= (49,200 + 10,000) - 40,000 \\ &= \text{RM}19,200 \end{aligned}$$

Therefore, it is worthwhile for Thompson to conduct a market survey as it is only cost 10,000 (<19,000)